

DIGITAL LOGIC & DESIGN (EE-227)

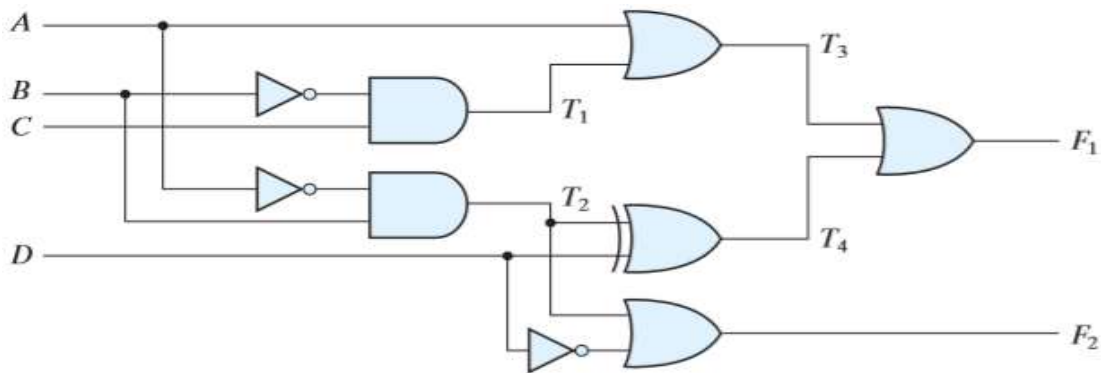
ASSIGNMENT #3

ID: _____

NAME: _____

SECTION: _____

1. Consider the following combinational circuit



- Derive the Boolean expressions for the outputs **F1** and **F2** as a function of the four inputs.
- List the truth table with 16 binary combinations of the four input variables. Then list the binary values for **T1** through **T4** and outputs **F1** and **F2** in the table.
- Simplify Boolean expressions of **F1** and **F2** using Boolean algebra and K-maps.
- List the truth table for simplified expression **F1** and **F2** and compare both truth tables

EXPRESSIONS

TRUTH TABLE

	A	B	C	D	T1	T2	T3	T4	F1	F2
0	0	0	0	0						
1	0	0	0	1						
2	0	0	1	0						
3	0	0	1	1						
4	0	1	0	0						
5	0	1	0	1						
6	0	1	1	0						
7	0	1	1	1						
8	1	0	0	0						
9	1	0	0	1						
10	1	0	1	0						
11	1	0	1	1						
12	1	1	0	0						
13	1	1	0	1						
14	1	1	1	0						
15	1	1	1	1						

ASSIGNMENT # 3 [EE-227]

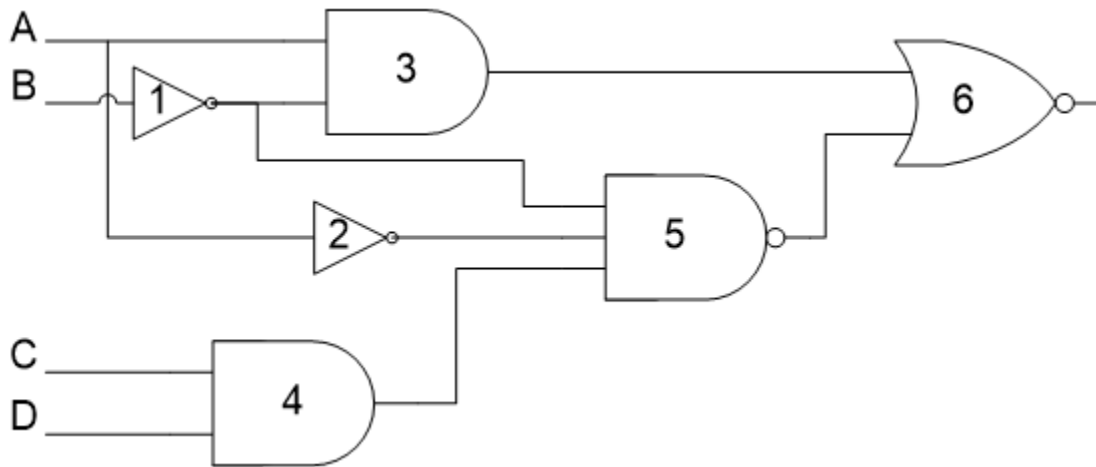
SIMPLIFICATION OF EXPRESSIONS:

TRUTH TABLE

	A	B	C	D	F1	F2		A	B	C	D	F1	F2
0	0	0	0	0			8	1	0	0	0		
1	0	0	0	1			9	1	0	0	1		
2	0	0	1	0			10	1	0	1	0		
3	0	0	1	1			11	1	0	1	1		
4	0	1	0	0			12	1	1	0	0		
5	0	1	0	1			13	1	1	0	1		
6	0	1	1	0			14	1	1	1	0		
7	0	1	1	1			15	1	1	1	1		

COMPARE RESULTS:

2. Consider the following combinational circuit



- Derive the Boolean expressions for the output **F1** as a function of the four inputs.
- List the truth table with 16 binary combinations of the four input variables. Then list the binary values output **F1** in the table.
- Simplify Boolean expressions of **F1** using Boolean algebra and K-maps.
- List the truth table for simplified expression **F1** and compare both truth tables.

EXPRESSION

TRUTH TABLE

	A	B	C	D	F1		A	B	C	D	F1	
0	0	0	0	0		8	1	0	0	0		
1	0	0	0	1		9	1	0	0	1		
2	0	0	1	0		10	1	0	1	0		
3	0	0	1	1		11	1	0	1	1		
4	0	1	0	0		12	1	1	0	0		
5	0	1	0	1		13	1	1	0	1		
6	0	1	1	0		14	1	1	1	0		
7	0	1	1	1		15	1	1	1	1		

ASSIGNMENT # 3 [EE-227]

SIMPLICATION OF EXPRESSION:

TRUTH TABLE

	A	B	C	D	F1		A	B	C	D	F1
0	0	0	0	0		8	1	0	0	0	
1	0	0	0	1		9	1	0	0	1	
2	0	0	1	0		10	1	0	1	0	
3	0	0	1	1		11	1	0	1	1	
4	0	1	0	0		12	1	1	0	0	
5	0	1	0	1		13	1	1	0	1	
6	0	1	1	0		14	1	1	1	0	
7	0	1	1	1		15	1	1	1	1	

COMPARE RESULTS:

ASSIGNMENT # 3 [EE-227]

3. Design a combinational circuit that converts a **three-bit Binary number** to a **3-bit Gray code**.

NOTE: Implement the circuit with exclusive-OR gates.

TABLE

BINARY				GRAY CODE		
	A	B	C	G2	G1	G0
0	0	0	0			
1	0	0	1			
2	0	1	0			
3	0	1	1			
4	1	0	0			
5	1	0	1			
6	1	1	0			
7	1	1	1			

SIMPLIFICATION:

EXPRESSIONS:

ASSIGNMENT # 3 [EE-227]

4. Design a combinational circuit that converts a **3-bit Gray code** to a **three-bit Binary number**.

NOTE: Implement the circuit with exclusive-OR gates.

TABLE

GRAY CODE				BINARY		
	G2	G1	G0	A	B	C
0				0	0	0
1				0	0	1
2				0	1	0
3				0	1	1
4				1	0	0
5				1	0	1
6				1	1	0
7				1	1	1

SIMPLIFICATION:

EXPRESSIONS:

ASSIGNMENT # 3 [EE-227]

5. Design a combinational circuit that converts a bit four binary number to a four-bit Gray code.

BINARY					GRAY CODE			
	A	B	C	D	G3	G2	G1	G0
0	0	0	0	0				
1	0	0	0	1				
2	0	0	1	0				
3	0	0	1	1				
4	0	1	0	0				
5	0	1	0	1				
6	0	1	1	0				
7	0	1	1	1				
8	1	0	0	0				
9	1	0	0	1				
10	1	0	1	0				
11	1	0	1	1				
12	1	1	0	0				
13	1	1	0	1				
14	1	1	1	0				
15	1	1	1	1				

SIMPLIFICATION:

EXPRESSIONS:

ASSIGNMENT # 3 [EE-227]

6. Design a combinational circuit that converts a BCD to an 8,4, -2, -1 code.

BCD CODE					8,4, -2, -1 CODE			
	A	B	C	D	8	4	-2	-1
0	0	0	0	0				
1	0	0	0	1				
2	0	0	1	0				
3	0	0	1	1				
4	0	1	0	0				
5	0	1	0	1				
6	0	1	1	0				
7	0	1	1	1				
8	1	0	0	0				
9	1	0	0	1				
10	1	0	1	0				
11	1	0	1	1				
12	1	1	0	0				
13	1	1	0	1				
14	1	1	1	0				
15	1	1	1	1				

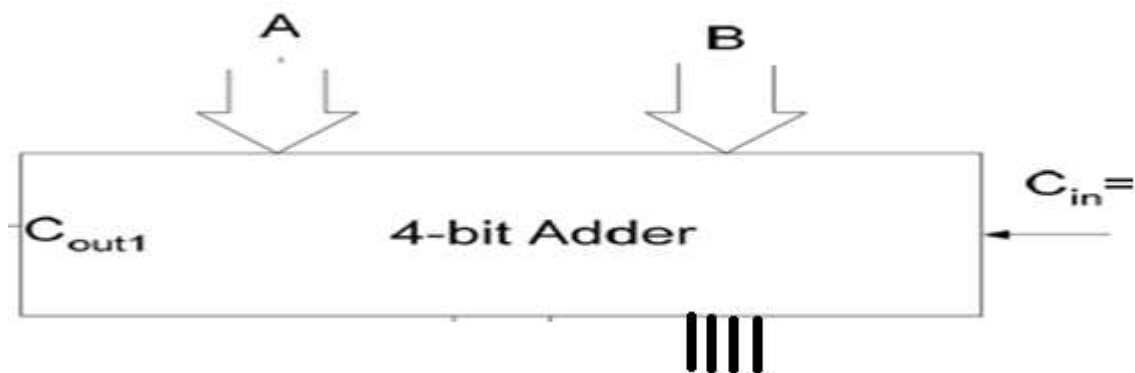
SIMPLIFICATION:

EXPRESSIONS:

7. Design **4-bit adder** using just **Half adders**? Only circuit required?

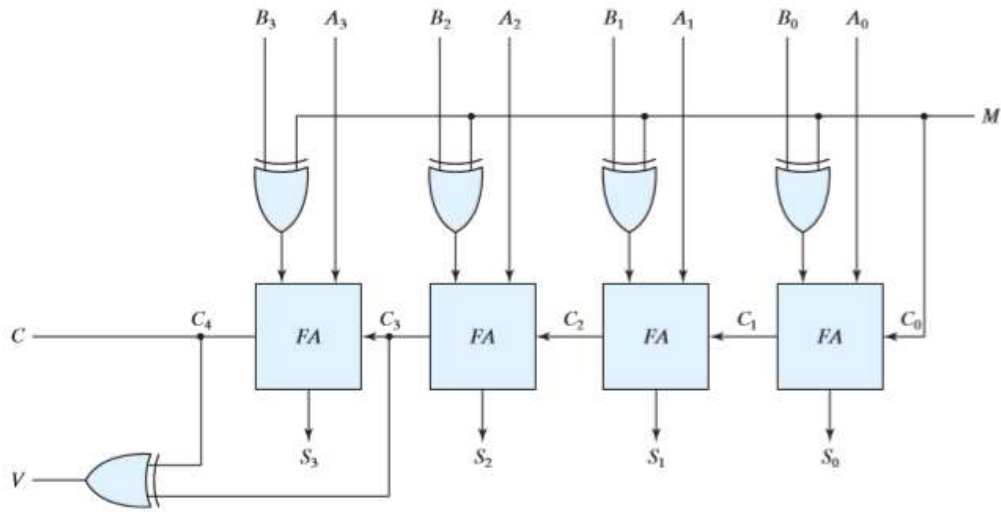
8. Convert following **4-bit adder** to 4-bit subtractor circuit? So it could perform operation **A-B**?

Note: B could be represented in 2's complement



ASSIGNMENT # 3 [EE-227]

9. Perform unsigned Addition $A+B$ where $A=9$ and $B=7$ using Following is **4-adder/subtractor** circuit? Also write short note on working of **C** and **V** bits for signed and unsigned addition/subtraction



ASSIGNMENT # 3 [EE-227]

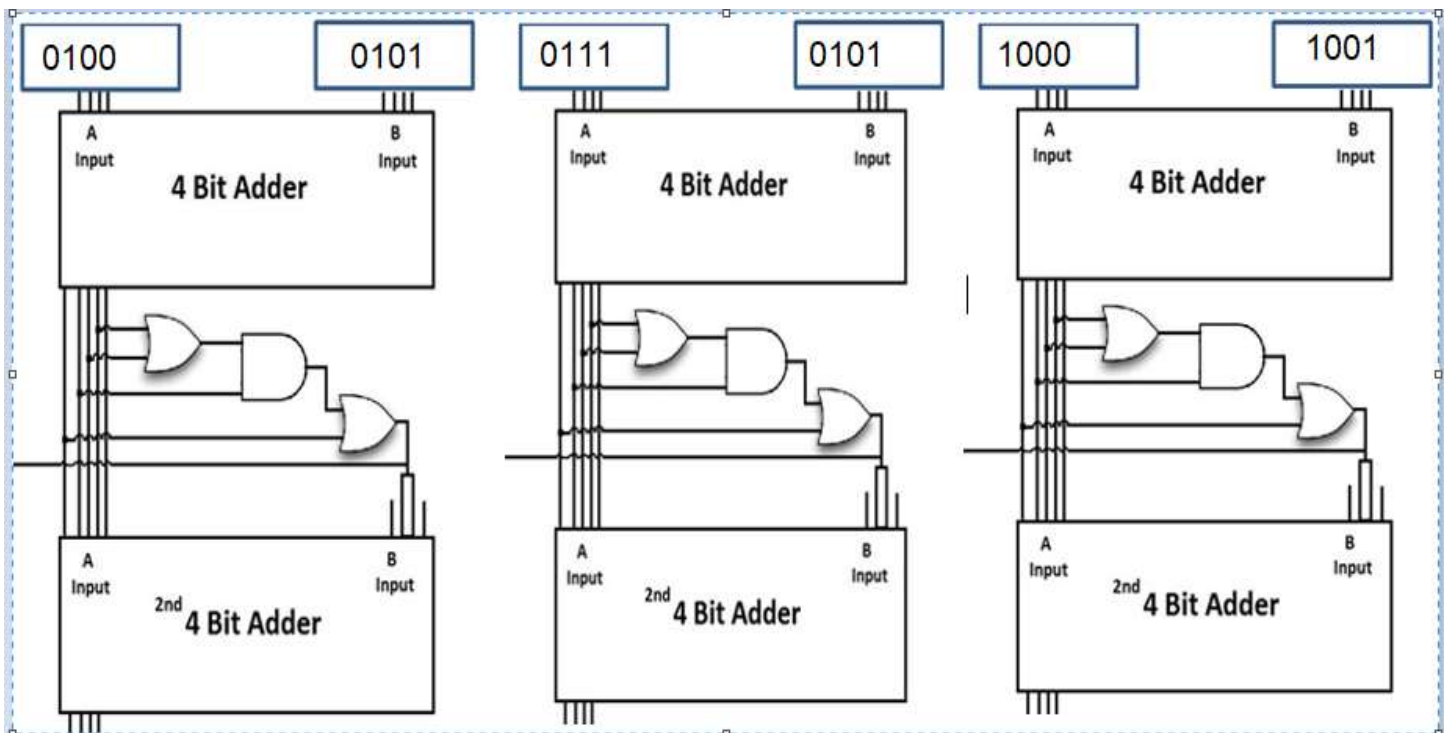
10. Design Circuit that accepts 4-bit BCD number as input and Detects BCD is valid/invalid?

	A	B	C	D	F	A	B	C	D	F
0	0	0	0	0		8	1	0	0	0
1	0	0	0	1		9	1	0	0	1
2	0	0	1	0		10	1	0	1	0
3	0	0	1	1		11	1	0	1	1
4	0	1	0	0		12	1	1	0	0
5	0	1	0	1		13	1	1	0	1
6	0	1	1	0		14	1	1	1	0
7	0	1	1	1		15	1	1	1	1

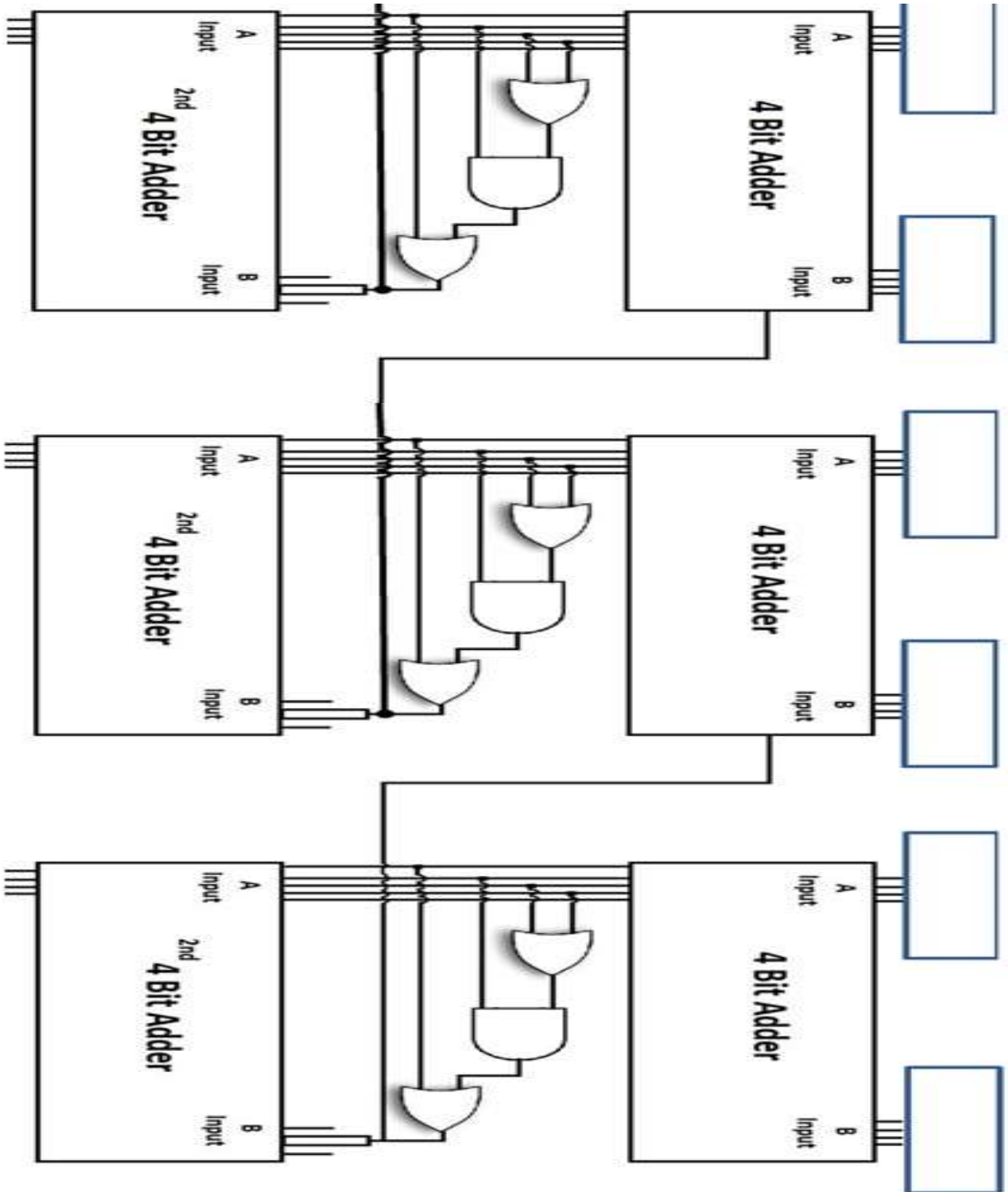
SIMPLIFICATION:

EXPRESSIONS:

11. Perform **BCD Addition** using following circuit? BCD numbers are given in boxes on top?



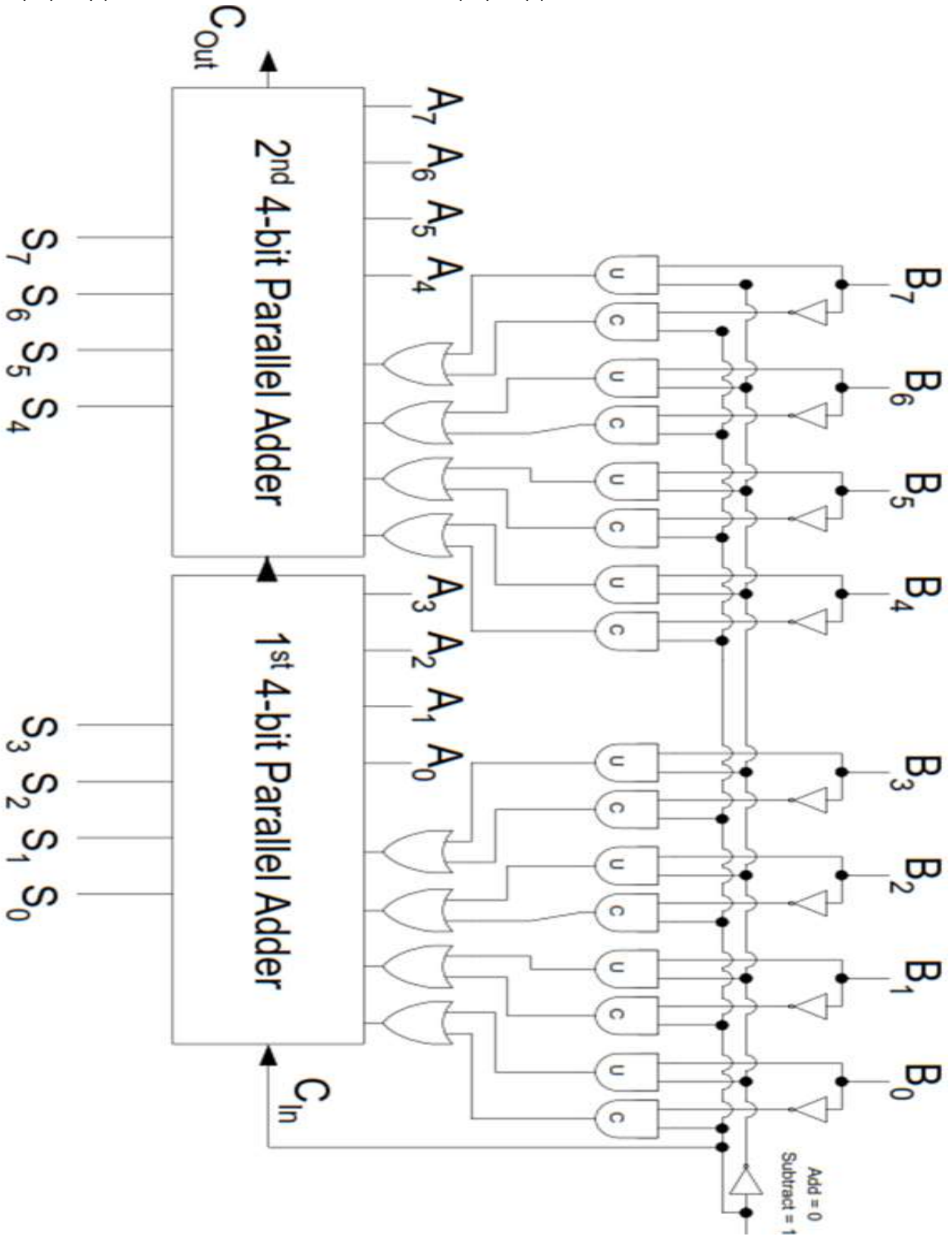
12. Perform BCD addition $(258)_{\text{BCD}} + (449)_{\text{BCD}}$ using following circuit?



13. Implement the binary arithmetic operation (A-B), using the following 8-bit addition/Subtraction circuit?

$$A = (52)_{10} = (?)_2$$

$$B = (25)_{10} = (?)_2$$



ASSIGNMENT # 3 [EE-227]

14. Design a single bit comparator from the following table? Write expression for $A=B$, $A<B$ and $A>B$? Design circuit from expression using logic gates?

TABLE:

INPUT		OUTPUT		
A	B	A>B	A==B	A<B
0	0			
0	1			
1	0			
1	1			

EXPRESSIONS

$F_{A=B} =$

$F_{A>B} =$

$F_{A<B} =$

15. Write expression for **5-bit** comparator? (Use section 4.6(Morris Mono) to design 5-bit comparator)

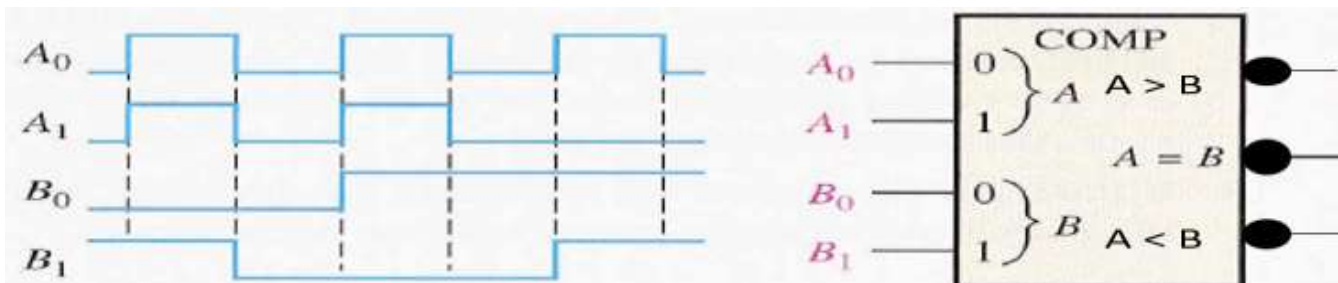
EXPRESSIONS

$F_{A=B} =$

$F_{A>B} =$

$F_{A<B} =$

16. Apply the Waveform give below to the comparator circuit. Determine the output waveforms of $A=B$, $A<B$, $A>B$ which are active low enable.



WAVEFORMS

$F_{A=B} =$

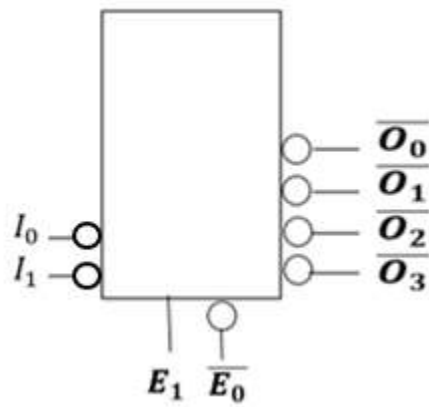
$F_{A>B} =$

$F_{A<B} =$

ASSIGNMENT # 3 [EE-227]

17. Design following **2x4 Decoder**? For reference see Section 4.9(figure 4.19)

BLOCK DIAGRAM



TABLE

INPUT				OUTPUT			
E_1	$\overline{E_0}$	$\overline{I_1}$	$\overline{I_0}$	$\overline{O_0}$	$\overline{O_1}$	$\overline{O_2}$	$\overline{O_3}$

EXPRESSIONS:

$$\overline{O_0} =$$

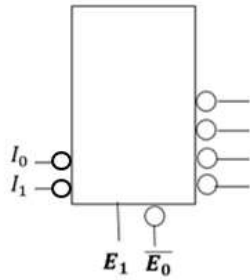
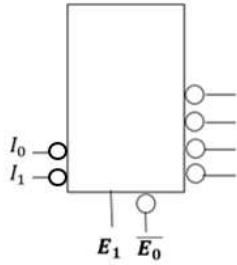
$$\overline{O_1} =$$

$$\overline{O_2} =$$

$$\overline{O_3} =$$

ASSIGNMENT # 3 [EE-227]

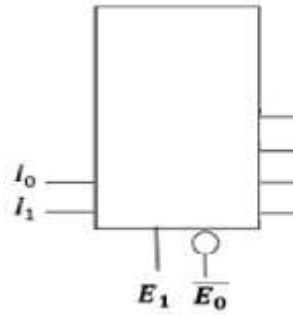
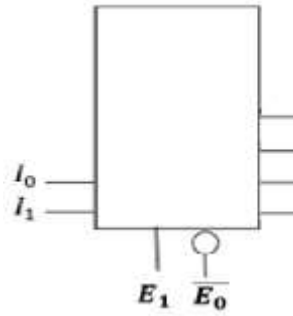
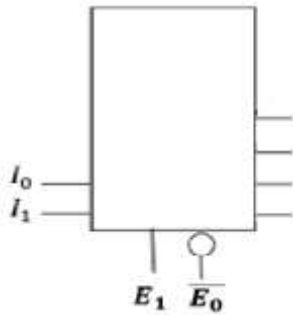
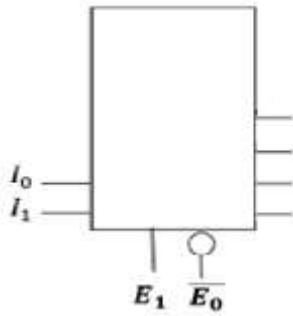
18. Use 2x4 decoder to design 3x8 decoder? Rename **OUTPUTS** and identify least and most decoder



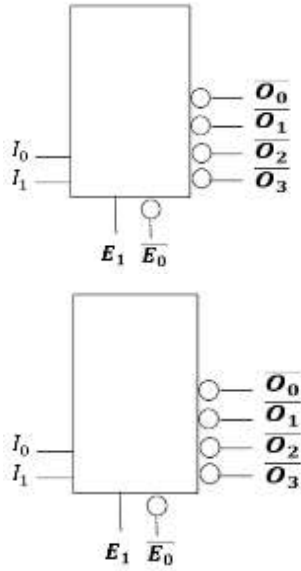
19. Use following 2x4 decoders to implement given function

a. $F(A,B,C) = \sum(0,1,2,4,6)$

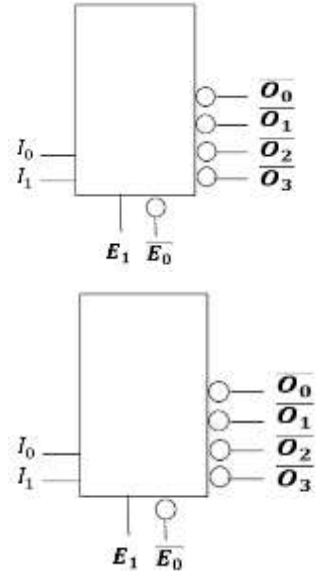
b. $F(A,B,C) = \prod(1,4,5,6)$



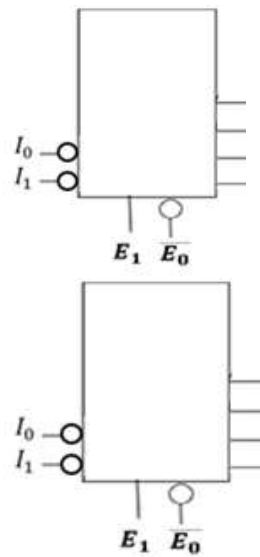
c. $F(A,B,C) = \sum(0,1,2,4,6)$



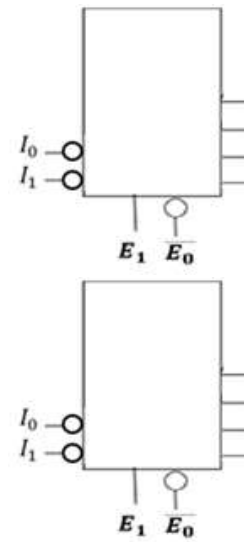
d. $F(A,B,C) = \prod(1,4,5,6)$



e. $F(A,B,C) = \sum(0,1,2,4,6)$

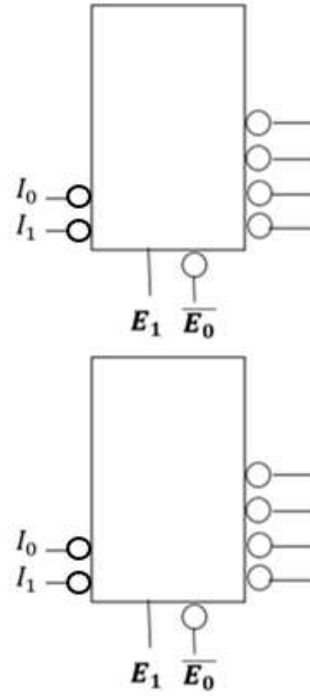
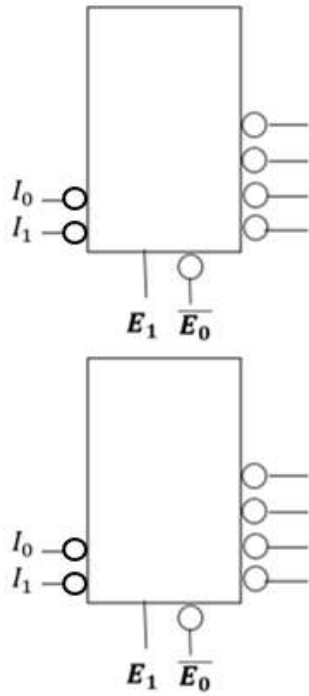


f. $F(A,B,C) = \prod(1,4,5,6)$

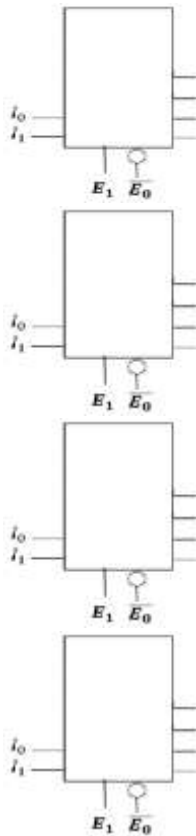


g. $F(A,B,C) = \sum(0,1,2,4,6)$

h. $F(A,B,C) = \prod(1,4,5,6)$



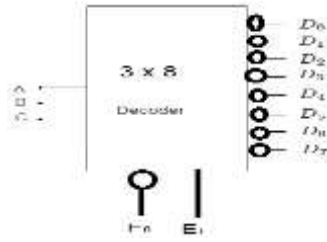
20. Use following **2x4 decoder** to construct **4 X 16 decoder**. You are not supposed to use any external gates?



ASSIGNMENT # 3 [EE-227]

21. Design following **3x8 Decoder**? For reference see Section 4.9(figure 4.18, table 4.6)

BLOCK DIAGRAM



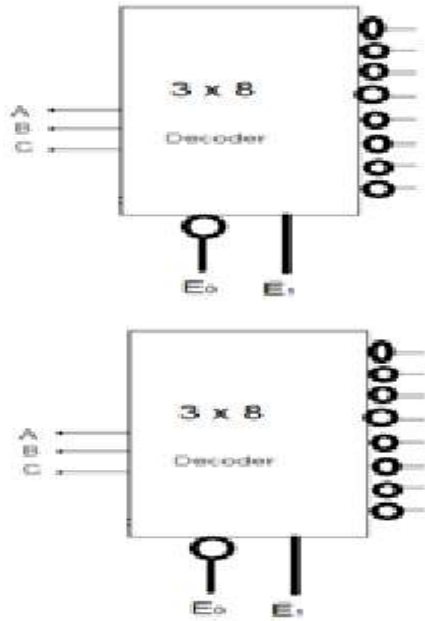
TABLE

INPUT					OUTPUT							
$\overline{E_0}$	E_1	A	B	C	$\overline{D_0}$	$\overline{D_1}$	$\overline{D_2}$	$\overline{D_3}$	$\overline{D_4}$	$\overline{D_5}$	$\overline{D_6}$	$\overline{D_7}$
X	X	X	X	X	1	1	1	1	1	1	1	1
X	X	X	X	X	1	1	1	1	1	1	1	1
		0	0	0								
		0	0	1								
		0	1	0								
		0	1	1								
		1	0	0								
		1	0	1								
		1	1	0								
		1	1	1								

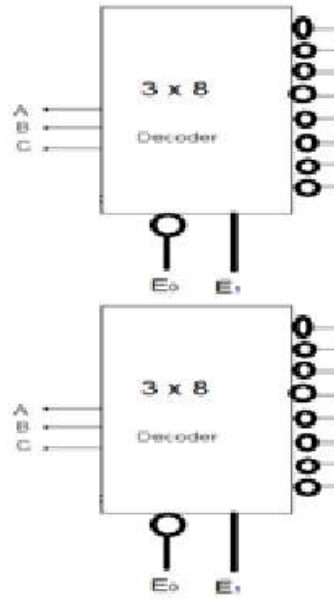
EXPRESSIONS

$\overline{D_0} =$	$\overline{D_1} =$	$\overline{D_2} =$
$\overline{D_3} =$	$\overline{D_4} =$	$\overline{D_5} =$
$\overline{D_6} =$	$\overline{D_7} =$	

22. Use above designed 3 X 8 decoder to design 4 X 16 decoder



23. Implement following function using two 3 X 8 decoder $F(A,B,C,D) = \sum(0, 3, 5, 7, 11, 15)$



24. Design 4X2 Low Priority Encoder? For reference see section 4.10(table 4.8)

TABLE

INPUT				OUTPUT		
D ₀	D ₁	D ₂	D ₃	A	B	Valid
0	0	0	0			
1	X	X	X			
0	1	X	X			
0	0	1	X			
0	0	0	1			

SIMPLIFICATION AND EXPRESSIONS

ASSIGNMENT # 3 [EE-227]

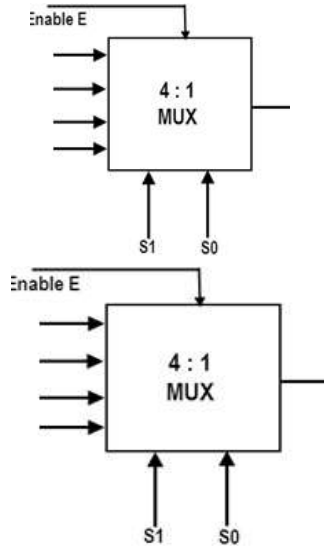
25. Design 8 X 3 Low Priority Encoder? For reference see slides Lec_17 uploaded on google classroom?

TABLE

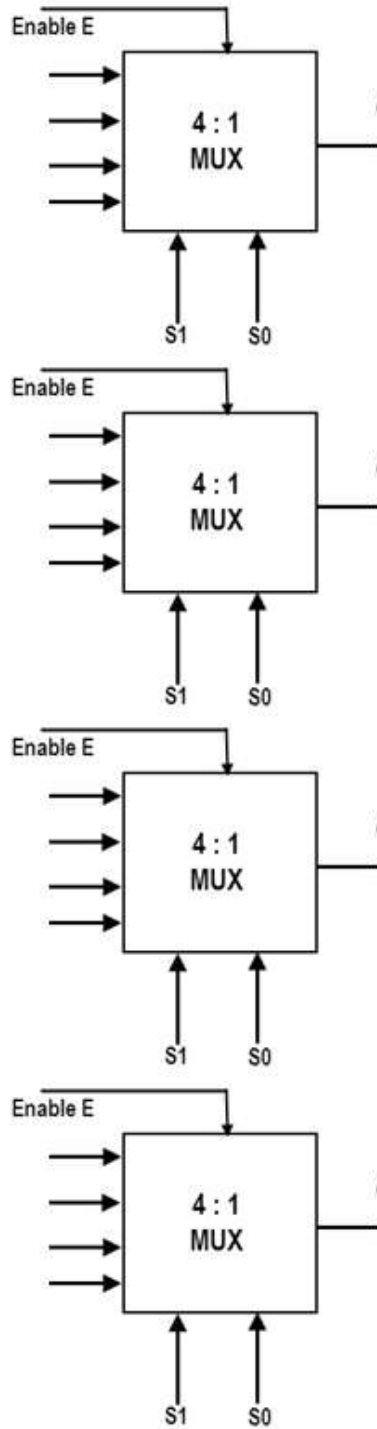
INPUTS							OUTPUTS		
I_0	I_1	I_3	I_4	I_5	I_6	I_7	O_0	O_1	O_2
1	X	X	X	X	X	X			
0	1	X	X	X	X	X			
0	0	1	X	X	X	X			
0	0	0	1	X	X	X			
0	0	0	0	1	X	X			
0	0	0	0	0	1	X			
0	0	0	0	0	0	1			

SIMPLIFICATION AND EXPRESSION

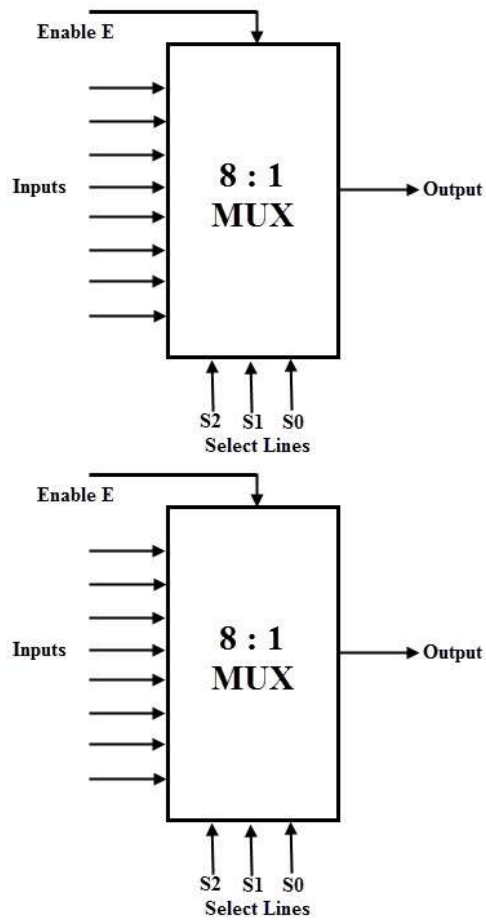
26. Use following 4 X 1 multiplexer to create 8 X 1 multiplexer?



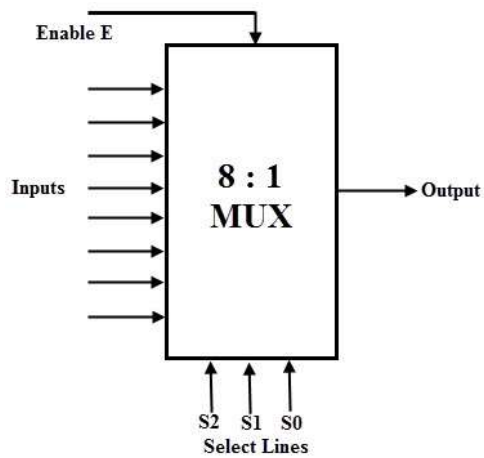
27. Use following 4 X 1 multiplexer to create 16 X 1 multiplexer?



28. Use following 8 X 1 multiplexer to create 16 X 1 multiplexer?

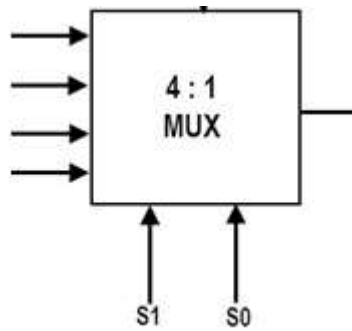


29. Implement given function using 8 X 1 multiplexer? $F(x, y, z) = \sum(0, 1, 2, 4, 5, 6)$



ASSIGNMENT # 3 [EE-227]

30. Implement given function using 4 X 1 multiplexer? $F(x, y, z) = \sum(0, 1, 2, 4, 5, 6)$



TABLE

X	Y	Z	F
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

31. Implement given function using 8 X 1 multiplexer? $F(w, x, y, z) = \sum(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$

